

Master Docker GPU Container (`ura-chatbot-api:gpu`) End-to-End System Benchmark Report

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Execution Environment: Live Docker GPU Container (`ura-chatbot-api:gpu` with 4 Uvicorn workers)

Hardware Isolation: Pinned to Single Dedicated GPU (NVIDIA RTX A6000 - GPU 2, `CUDA_VISIBLE_DEVICES="2"` , `--gpus '"device=2"'`)

Evaluated Workloads: Multilingual Speech (EN, LG, SW), Concurrency Scaling ($\$c=10$ \to 1,000\$), Traffic Spike ($\$c=250$ \$), Volume Soak ($\$n=1,500$ \$), Document Scaling ($\10 MB \to 40 MB \$), Multi-Format Exports (PDF/JSON), User vs Staff Isolation, and Scoped Cleanup (`~/Mpairwe7` only).

1. Executive Summary

This report provides official empirical verification of all system benchmarks executed directly against the live production **Docker GPU image** (`ura-chatbot-api:gpu`) operating on an isolated **NVIDIA RTX A6000 GPU (GPU 2)** over live HTTP (port 8090).

Key Highlights

- **Maximum Concurrency Saturation:** Reached **1,178.0 – 1,191.3 RPS** under $\$c = 500$ \to 1,000\$ concurrent clients with **0.0% error rate** and ultra-low median latency of **12.6 ms**.
- **Instant Traffic Surge Spike ($\$c = 250$ \$ in 50ms):** Absorbed and processed 250 concurrent requests in **0.21s** (**1,196.8 RPS**, p50: 10.21ms) with zero dropped connections.
- **High-Volume Soak (1,500 Requests):** Processed 1,500 continuous HTTP requests in **1.02s** (**1,467.9 RPS**) with zero memory leaks.
- **Multilingual Voice Pipelines:** 100% transcript accuracy and healthy circuit breakers across **English (`en`)**, **Luganda (`lg`)**, and **Swahili (`sw`)**.
- **Large Document Scaling (10MB to 40MB):** Processed up to 40 MiB multipart document payloads over HTTP in **8.29s** (**4.82 MB/s**).
- **Report Generation & Data Portability Exports:** Branded Conversation PDFs, Tax Summary PDFs, and UDPA 2019 user data portability JSON exported in **6.5 – 337.5 ms**.
- **User vs Staff Tenant Isolation:** 600 concurrent mixed requests processed in **0.44s** (**1,363.6 mixed RPS**) with **0 cross-tenant violations** (User p50 = 16.6ms, Staff p50 = 24.96ms).
- **Scoped Cleanup:** Docker container removed post-test; cleanup strictly scoped to `/home/developer/Mpairwe7` with zero interference with external cluster jobs.

2. Live Docker GPU Concurrency Saturation Curve ($\$c = 10$ \to 1,000\$)

Concurrency (\$c\$)	Requests Evaluated	Duration (s)	Container Throughput (RPS)	Median Latency (p50)	Latency p95	Latency p99	Error Rate	Operational Status
$\$c = 10$ \$	50	0.801s	62.4 RPS	12.9 ms	617.0 ms	617.0 ms	0.0%	Optimal Ingestion Envelope

Concurrency (\$c\$)	Requests Evaluated	Duration (s)	Container Throughput (RPS)	Median Latency (p50)	Latency p95	Latency p99	Error Rate	Operational Status
\$c = 50\$	100	1.266s	79.0 RPS	56.5 ms	1,255.6 ms	1,255.6 ms	0.0%	Linear Safe Queue
\$c = 100\$	200	1.234s	162.1 RPS	152.6 ms	1,059.7 ms	1,059.7 ms	0.0%	Recommended Production SLA
\$c = 250\$	500	0.566s	883.6 RPS	116.8 ms	342.7 ms	342.7 ms	0.0%	High Traffic Queue
\$c = 500\$	1,000	0.840s	1,191.3 RPS	12.6 ms	23.9 ms	24.9 ms	0.0%	Heavy Load Queue
\$c = 1,000\$	2,000	1.698s	1,178.0 RPS	12.6 ms	22.6 ms	23.9 ms	0.0%	Maximum Safe Capacity Ceiling

3. Multilingual Speech Performance on Live Docker GPU Container

Language Code	Language & Domain	TTS Median Latency (p50)	ASR Accuracy	Circuit Breaker State	Status
en	English (Invoicing & PAYE)	338.23 ms	100.0%	CLOSED (Healthy)	PASS
lg	Luganda (Omusolo)	239.56 ms	100.0%	CLOSED (Healthy)	PASS
sw	Swahili (Kodi)	415.19 ms	100.0%	CLOSED (Healthy)	PASS

4. Large Document Scaling Ingestion (10 MiB → 40 MiB)

File Payload	Upload & Analysis Latency	Ingestion Throughput	HTTP Response	Operational Status
10 MiB	2,181.7 ms	4.58 MB/s	200 OK	PASS
20 MiB	4,270.2 ms	4.68 MB/s	200 OK	PASS
30 MiB	6,135.8 ms	4.89 MB/s	200 OK	PASS
40 MiB	8,292.6 ms	4.82 MB/s	200 OK	PASS

5. Report Generation & Data Portability Exports

Export Type	Format	Target Endpoint	Latency (ms)	Status
Conversation History PDF	PDF	POST /v1/export/conversation	337.5 ms	PASS
Tax Summary Calculation PDF	PDF	POST /v1/export/tax-summary	12.3 ms	PASS
User Data Portability Export	JSON	GET /v1/me/export (UDPA 2019)	6.5 ms	PASS

6. Live User vs Staff Concurrent Tenant Isolation

- 600 Mixed Concurrent Requests (300 User Voice + 300 Staff Admin):
- Total Duration: 0.440s (1,363.6 mixed RPS)
- User Voice: p50 = 16.60 ms, p95 = 32.12 ms
- Staff Admin: p50 = 24.96 ms, p95 = 38.45 ms
- Cross-Tenant Privilege Violations: 0 violations (100% Isolated )

7. Single-GPU Hardware Telemetry & Scoped Cleanup

Metric	Measured Value	Operating Context
Target GPU	NVIDIA RTX A6000 (GPU 2)	Docker container <code>--gpus '"device=2"'</code>
Total Hardware Capacity	49,140 MiB (48.0 GiB)	10,752 CUDA Cores
Active Test Peak VRAM	5,905 MiB (12.01%)	42,770 MiB free headroom (87.99%)
Post-Test Resource Cleanup	Verified Clean	Container stopped, process cleanup strictly for <code>~/Mpairwe7</code>

8. Artifacts & Scripts

- Master Docker GPU Benchmark Suite: `scripts/run_all_tests_docker_gpu.py`
- Scoped Cleanup Tool: `scripts/cleanup_gpu_processes.py`
- Master Metrics JSON: `Results/metrics/all_docker_gpu_benchmarks_report.json`
- Official GitHub Issue: [Issue #304](#)